HARD · INFORMATION AND IDEAS · ANSWER KEY

Inferences

01

10 answers · Rationale-rich · Teach from doc

Q1 B

B. it does not allow for multiple castings of the same burrow over time.

Choice B is the best answer. Since resin casting "inevitably requires destroying the burrow," it would be impossible to make multiple castings of the same burrow over time.

Choice A is incorrect. The passage doesn't discuss size requirements for completing resin casting, so there's no basis for this inference. Choice C is incorrect. The passage never mentions how long the casting process takes nor how long *A. bellulus* takes to construct a borrow, so there's no basis for this inference. Choice D is incorrect. The passage never states that recovering the model distorts the resin's shape, only that it destroys the burrow. Therefore, there's no basis for this inference.

Q2 A

A. will likely continue to be used primarily to observe objects outside the solar system.

Choice A is the best answer. The HST will operate until at least 2030, but it's only observing stuff inside our solar system 6% of the time. If we could get a different telescope to observe stuff inside our solar system 100% of the time and take more extensive images of certain things, then the HST could continue to be used mainly for observing stuff outside the solar system.

Choice B is incorrect. This inference is too strong to be supported by the text. Even if the new telescope is deployed, the HST might still be used as it's being used now. Based on the text, the new telescope would just be used for more extensive and long-term imaging of solar system bodies, which doesn't necessarily overlap with the HST. Choice C is incorrect. This inference isn't supported. The text never mentions the possibility of modifying the HST, so there is no basis to make this inference. Rather, the researchers suggest using a different telescope to more closely observe certain objects. Choice D is incorrect. This inference is too strong to be supported. The text doesn't give us enough info to assume that the HST lacks any particular sensors.

Q3

D

D. rather than undergoing changes resulting in the eventual elimination of a head region in their radial body plan, as previously assumed, sea stars' morphology evolved to completely lack a trunk and consist primarily of a head region.

Choice D is the best answer because it most logically completes the text's discussion of the morphology (form and structure) of sea stars, a type of echinoderm. The text indicates that echinoderms have radially symmetrical body plans (symmetrical around a central point, usually in the form of a star), whereas most animals have bilaterally symmetrical body plans (symmetrical along an axis running from head to tail through a trunk). According to the text, sea stars are unusual echinoderms because, despite their radial body plan, they descended from known bilateral ancestors. This shift in body plan was thought to be a process of losing the genetic markers associated with the head region. The text explains that by comparing the genes of one sea star species (*P. miniata*) to those of a close relative, the acorn worm, researchers determined that instead, anterior (head) genes are active across the sea star's entire body, posterior (tail) genes are active in limited, peripheral locations of the body, and no trunk-related genes are active. This finding strongly suggests that, rather than becoming "headless" as they evolved from a bilateral ancestor, sea stars developed a body plan consisting almost entirely of a head region with a minimal tail region and no trunk region present.

Choice A is incorrect because the text doesn't identify how any particular region of sea stars' bodies influences the layout of sea stars' radial symmetry. Moreover, the text indicates that the radial symmetry of echinoderms is "usually starlike," not that a starlike layout distinguishes sea stars from other echinoderms. Choice B is incorrect because the text doesn't suggest that the idea that sea stars evolved from an ancestor with bilateral symmetry is incorrect (describing the bilateral origin as "known") and doesn't address any body plans other than those with radial or bilateral symmetry. The text strongly suggests that rather than revealing something about sea stars' origin, Formery et al.'s findings contradict the assumption that the current body plan of sea stars is "headless." Choice C is incorrect because the text suggests that Formery et al. were able to make determinations about *P. miniata* sea stars' body plan based on the comparability of genetic markers between *P. miniata* and *S. kowalevskii* acorn worms. The text indicates only that little or no activity was observed in certain types of genes associated with body development in *P. miniata*, not that those genes turned out to largely differ from body-development genes in *S. kowalevskii*.

Q4

D

D. the costs of writing, editing, and designing a book were less affected by the technologies used than were the costs of manufacturing and distributing a book.

Choice D is the best answer because it presents the conclusion that most logically follows from the text's discussion of how digital technologies affected the process of book creation. The text explains that in the late 20th and early 21st centuries digital technologies lowered book production costs most significantly in manufacturing and distribution. The text goes on to point out that authoring, editing, and book design are distinct steps in the process that occur before manufacturing and distribution. Because the savings connected to digital technologies have been most significant in manufacturing and distribution, it's reasonable to infer that those technologies had less of an effect on writing, editing, and designing books.

Choice A is incorrect because the text focuses on lowered book production costs that occur after authoring has taken place; there's no indication in the text whether digital technologies made writing and publishing lengthy books easier. Choice B is incorrect. Although it's logical to conclude that customers would expect the cost of books to decline if production costs have declined, the text doesn't address customer expectations for the cost of books or any other consumer goods. Choice C is incorrect because the text focuses broadly on how digital technologies have affected the cost of the publishing process; it doesn't address the kinds of books being published or how many copies are printed.

Q5

A

A. risk judging Tanner's painting by standards that may not be historically appropriate.

Choice A is the best answer. It most logically completes the text. The text argues that Tanner and his contemporaries may have been unfamiliar with modern beliefs and values. This suggests that scholars who attribute those modern values to Tanner's painting are risking judging the painting by standards that are not historically accurate.

Choice B is incorrect. It doesn't logically complete the text. The text argues that Tanner AND his contemporaries may have been unfamiliar with modern views. It never suggests that Tanner's views were different from his contemporaries' views. Choice C is incorrect. It doesn't logically complete the text. The text never suggests that scholars should analyze Tanner's political activity instead of his painting. Rather, the text argues that Tanner and his contemporaries may have been unfamiliar with modern beliefs and values. Choice D is incorrect. It doesn't logically complete the text. The text never suggests that Tanner wanted to critique his contemporaries with his painting. Rather, the text argues that Tanner AND his contemporaries may have been unfamiliar with modern beliefs and values.

Q6

D

D. because the responses of bumblebees and other wild bees to environmental threats are not always comparable, researchers need to exercise caution when extrapolating information about wild-bee population declines from bumblebees.

Choice D is the best answer because it most logically completes the text's discussion of wild-bee declines. The text establishes that bumblebees and other wild bees have been experiencing population collapse as a result of habitat destruction, climate variation, and other factors. The text then indicates that because bumblebees are very extensively researched, ecologists rely heavily on findings about them to understand wild-bee declines in general. However, the text then introduces Ghisbain's observation that unlike most other wild-bee genera, bumblebees have certain traits (social behaviors and dietary generalism) linked to increased resilience to certain environmental changes. In other words, bumblebees aren't necessarily representative of wild bees as a whole because they are likely more tolerant of some pressures. Therefore, it logically follows that Ghisbain would urge researchers to exercise caution when using bumblebee data to draw conclusions about other wild-bee population declines, since bumblebees and other wild bees don't always respond comparably to environmental threats.

Choice A is incorrect because the text doesn't indicate that climate variation is more of a threat to bumblebees than habitat destruction is or that it is a bigger threat to bumblebees than it is to other wild bees. The text simply indicates that climate variation and habitat destruction are among the factors that have caused population collapses for bumblebees and other wild-bee genera. Choice B is incorrect. As it is presented in the text, Ghisbain notes that bumblebees may respond to environmental changes differently than many other wild-bee genera do; this suggests that bumblebees shouldn't be treated as representative of all wild bees, but there's no reason to think that Ghisbain would go so far as to assert that bumblebee data shouldn't be used to draw conclusions about other wild-bee genera that are similar to bumblebees in their dietary and social traits, even if there are relatively few. The text suggests caution in extrapolation, not a complete rejection of using bumblebee research. Choice C is incorrect. Although the text indicates that dietary generalism is linked to increased resilience to specific environmental changes (that is, to some but not all environmental stresses), it doesn't suggest that bumblebees and other bees with this trait are less likely than bees with specialized diets to experience major population changes. In fact, the text makes it clear that like many other wild-bee genera—most of which don't display dietary generalism—bumblebees are already experiencing population collapse.

Q7

C

C. ELF3 enables *A. thaliana* to respond to increased temperatures.

Choice C is the best answer because it most logically completes the text's discussion of accelerated flowering in *A. thaliana* plants. The text indicates that *A. thaliana* plants show accelerated flowering at high temperatures. To investigate the mechanism for this accelerated flowering, biologists replaced the ELF3 protein in one group of *A. thaliana* plants with a similar protein found in another plant species that doesn't show accelerated flowering. The team then compared these modified plants to *A. thaliana* plants that retained their original ELF3 protein. The text states that the two samples of plants showed no difference in flowering at 22° Celsius, but at 27° Celsius the unaltered plants with ELF3 showed accelerated flowering while the plants without ELF3 didn't. If accelerated flowering at the higher temperature occurred in the *A. thaliana* plants with ELF3 but not in the plants without the protein, then ELF3 likely enables *A. thaliana* to respond to increased temperatures.

Choice A is incorrect because the text doesn't mention whether any plants other than *A. thaliana* and stiff brome show temperature-sensitive flowering, so there is no support for the idea that this type of flowering is unique to *A. thaliana*. Choice B is incorrect because the text discusses the effects of ELF3 and not the production of it. There's nothing in the text to suggest that the amount of ELF3 in *A. thaliana* varies with temperature. Choice D is incorrect. While the text states that there was no difference in the flowering of modified and unmodified *A. thaliana* plants at 22° Celsius, there's no suggestion that *A. thaliana* only begins to flower at 22° Celsius; the text doesn't mention a specific temperature threshold required for *A. thaliana* flowering.

Q8

B

B. need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.

Choice B is the best answer. The text describes a study examining the relationship between a species of parasitic tapeworm, *A. brevis*, and its host insect, the *T. nylanderi* ant. According to the text, researchers were surprised to find that the tapeworm extends the life of its ant host, rather than reducing it. The text goes on to state that the infected ants end up doing less work to sustain the colony and that as a result, the uninfected ants take on the infected ants' share of labor in addition to caring for them in their infected state. The study's researchers also observed that the uninfected ants have shorter lifespans than expected. If the infected ants, who are doing less work in the colony, have longer lifespans, it can be inferred that the less an ant works, the longer it will live. The opposite of this statement can also be inferred: the more an ant works, the shorter its life. So, since the workload within the colony is being redistributed so that the infected ants work less while the uninfected ants work more (as they take on the neglected duties of the infected ants and also care for those ants), then it can be inferred that the lifespans of the uninfected ants are shortened because the need to compensate for reduced contributions within the colony while also caring for infected workers is burdensome to the uninfected workers.

Choice A is incorrect because the text does not indicate how *A. brevis* is transmitted to the ants or assert that uninfected ants are more likely to be directly exposed to *A. brevis* while caring for infected ants. Choice C is incorrect because the text makes no mention of the relative abilities of infected and uninfected ants to escape predators: in fact, predators are not mentioned in the text at all. Choice D is incorrect because the text does not supply any information about the average lifespans of the ants in colonies without parasitic activity; the text only indicates factors that lengthen and shorten the lifespans of ants in parasitized colonies.

Q9 B

B. keep the amount of silver in Sidonian coins consistent with that in coins minted in 367 BCE but decrease their weight.

Choice B is the best answer because it most logically completes the text's discussion of Sidonian coins. As the text explains, researchers determined that Sidonian coins were made of silver and copper and that from 450 BCE to 367 BCE, the percentage of silver in each coin decreased from 98% to 74.2% while the percentage of copper increased from 1% to 24.7%. The text indicates that because the coins containing less than 80% silver weren't considered suitable for trade (suggesting that copper was less valuable than silver) and looked different from coins containing more silver, the researchers suspect there was a serious loss in confidence in the currency in Sidon in 367 BCE when the copper content was high. It's reasonable to assume that it wasn't possible to boost confidence simply by devoting a greater amount of valuable silver to the currency, since Sidon was under significant and ongoing financial pressure; however, keeping the total amount of silver the same and reducing the amount of copper in the coins would have resulted in smaller coins with a higher percentage of silver. Therefore, it makes sense to suggest that Abd'aštart I (the ruler after 367 BCE) likely restored confidence in the currency by deciding to keep the amount of silver in Sidonian coins consistent with that in coins minted in 367 BCE but to decrease the coins' weight.

Choice A is incorrect because the text conveys that a crisis in confidence in the currency of Sidon likely occurred around 367 BCE because the percentage of silver in coins had fallen below 80% (presumably because Sidon's financial pressures meant that less silver was available for currency), making the coins unsuitable for trade. Thus, announcing that the threshold for the percentage of silver in coins would be raised—that is, that coins would need to contain even more than 80% silver to be suitable for trade—likely would have worsened the crisis rather than relieved it. Choice C is incorrect because the text strongly suggests that a crisis in confidence in the currency of Sidon was caused by the proportion of silver to copper in the coins in 367 BCE, with 74.2% being too little silver for the coins to be considered suitable for trade; therefore, it's unlikely that minting coins with a similar proportion of silver to copper (that is, still around 74.2% silver) would have restored confidence, even if the coins were heavier. Choice D is incorrect because the text gives no indication that funding the mining of more copper would have relieved a crisis in confidence in the currency of Sidon. The text establishes that Sidonian coins that visibly contained copper weren't considered suitable for trade, so Abd'aštart I wouldn't have wanted to add even more copper to them, and it's unclear how else copper mining would affect views of the currency.

Q10 A

A. those people who had the right to vote most likely had substantial holdings of French real estate.

Choice A is the best answer. We're told that people needed to pay "at least 300 francs in direct taxes" to be able to vote. We're also told that, while "relatively few people paid the tax on real estate," real estate taxes were both the main way people qualified to vote and the main source of revenue for the government. Based on this, we can infer that those who did qualify to vote likely had significant French real estate holdings.

Choice B is incorrect. The passage doesn't mention the voting habits of artisans and merchants nor any reduction in tax burdens on businesses, so there's no basis for this inference. Choice C is incorrect. Although we know that doors and windows were taxed during the Bourbon Restoration, we don't have enough information to infer if doors and windows increased after this time. Choice D is incorrect. Although we know that foreign investments were only minimally taxed, we don't have enough information to determine if those with significant foreign investments were unlikely to have voting rights. For example, it's possible that those with significant foreign investments were likely to also be people with significant domestic investments which they did pay taxes on, so we don't have the information necessary to make this inference.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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